AWS INTRODUCTION

Introduction to Amazon Web Services (AWS)

Amazon Web Services (AWS) is a cloud computing platform provided by Amazon. It offers on-demand computing services like servers, storage, databases, networking, security, and more — all over the internet.

Instead of buying and maintaining physical servers, you can rent resources from AWS and pay only for what you use.

Why AWS is Popular

- Pay-as-you-go pricing
- Highly scalable (increase/decrease resources anytime)
- Global infrastructure
- High availability & reliability
- Strong security features
- Supports DevOps, CI/CD, Docker, Kubernetes, etc.

Since you're working on Jenkins, Docker, EC2, and CI/CD setups, AWS is very useful for hosting and automating deployments.

AWS Global Infrastructure

AWS has:

- **Regions** – Physical locations around the world
- **Availability Zones (AZs)** – Multiple data centers inside a region
- **Edge Locations** – For fast content delivery (CDN)

Example:

- Mumbai Region → ap-south-1

Core AWS Services

Compute Services

- **Amazon EC2** – Virtual servers in the cloud

- **AWS Lambda** – Run code without managing servers
- **Amazon ECS** – Run Docker containers
- **Amazon EKS** – Managed Kubernetes

Storage Services

- **Amazon S3** – Store files, images, backups
- **Amazon EBS** – Storage for EC2
- **Amazon Glacier** – Long-term backup storage

Database Services

- **Amazon RDS** – MySQL, PostgreSQL, etc.
- **Amazon DynamoDB** – NoSQL database

Networking

- **Amazon VPC** – Private network in AWS
- **Elastic Load Balancing** – Distributes traffic
- **Amazon Route 53** – Domain and DNS management

DevOps & CI/CD

- **AWS CodePipeline**
- **AWS CodeBuild**
- **AWS CodeDeploy**

AWS Pricing Model

AWS follows:

- Pay only for what you use
- No upfront investment
- Free Tier available for beginners (12 months limited usage)

Who Uses AWS?

Many big companies use AWS:

- Netflix
- Airbnb
- Spotify
- NASA

Simple Example (Real Scenario)

If you want to:

- Host a Django app → Use **EC2 + RDS + S3**
- Deploy using Jenkins → Jenkins installed on EC2
- Run Docker containers → Use ECS or EC2 with Docker
- Host static website → Use S3

In One Line

AWS is a cloud platform that lets you build, deploy, and scale applications without buying physical servers.